Investigating Preprocessing Choices, Model Complexity, and Regularization Strategies in Seizure Prediction

Semester Major Assignment · Institute of Management Sciences, Peshawar

Submitted by: Hafiz Ullah · **Submitted to:** Dr. Adnan Amin

RESEARCH SCOPE

Project Objectives

Preprocessing Choices

Investigate how preprocessing order and techniques affect EEG signal quality and model input.

Model Complexity

Study the effects of feature dimensionality and regularization strength on generalization.

Class Imbalance

Evaluate strategies for handling imbalanced seizure vs. non-seizure class distributions.

Generalization Performance

Compare cross-dataset performance using Accuracy, F1-score, and PR-AUC metrics.

DATA FOUNDATION

Datasets Used

1

EEG Extracted Feature Dataset

Pre-extracted spectral and temporal features from scalp EEG recordings.

2

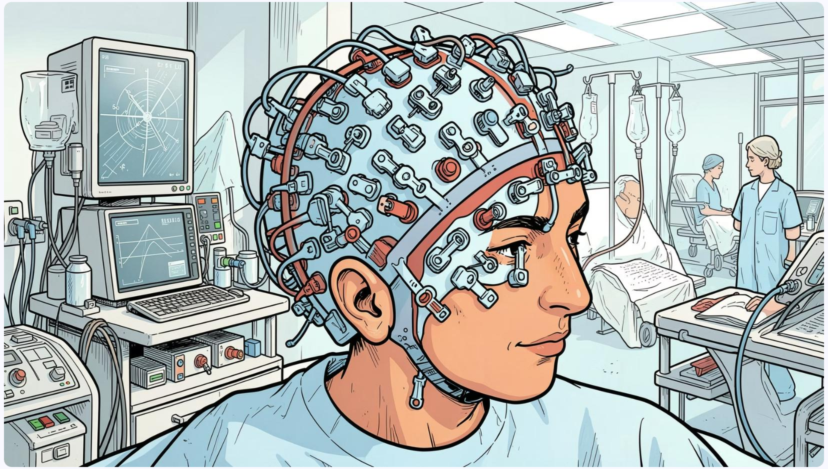
Multi-class Imbalanced Seizure Dataset

Multi-class labels with pronounced class imbalance for robustness testing.

3

CHB-MIT EEG Seizure Dataset

Publicly available clinical dataset with multiple patients and seizure events.



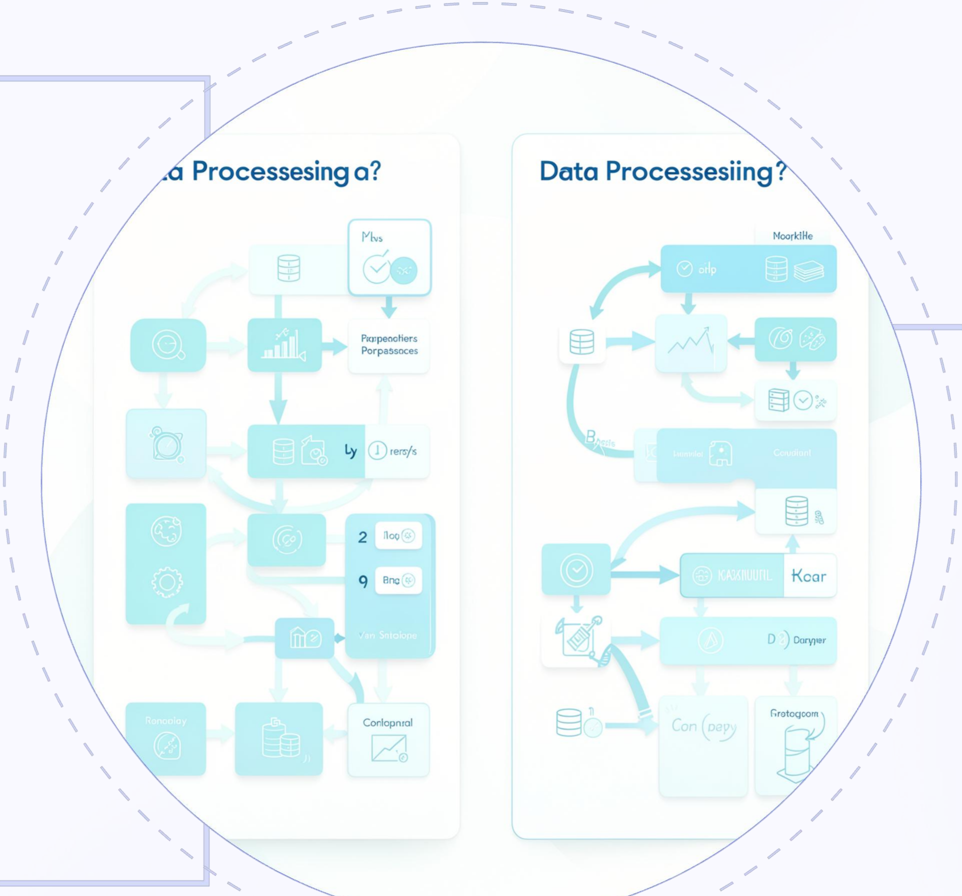
METHODOLOGY

Preprocessing Pipelines

Two distinct pipelines were designed to isolate the impact of preprocessing order on downstream classifier performance.

Pipeline A

Normalization → Noise Removal → Feature Selection



Pipeline B

Feature Extraction → Scaling → PCA

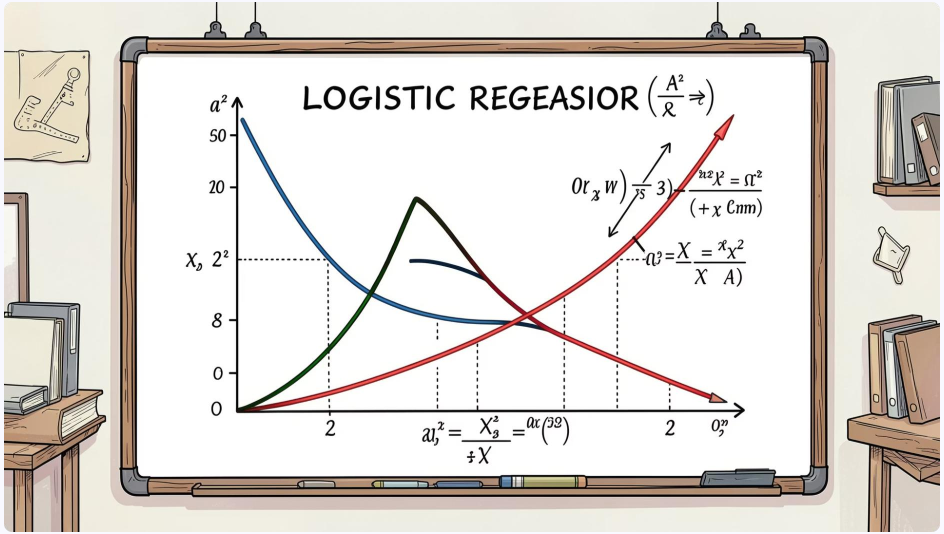
Compare

Evaluate classifier performance differences

Made with GAMMA

CORE MODEL

Baseline Logistic Regression



Logistic Regression serves as the interpretable baseline classifier, producing calibrated binary probability estimates for seizure vs. non-seizure classification.

Accuracy

Overall correctness across both classes.

F1-Score

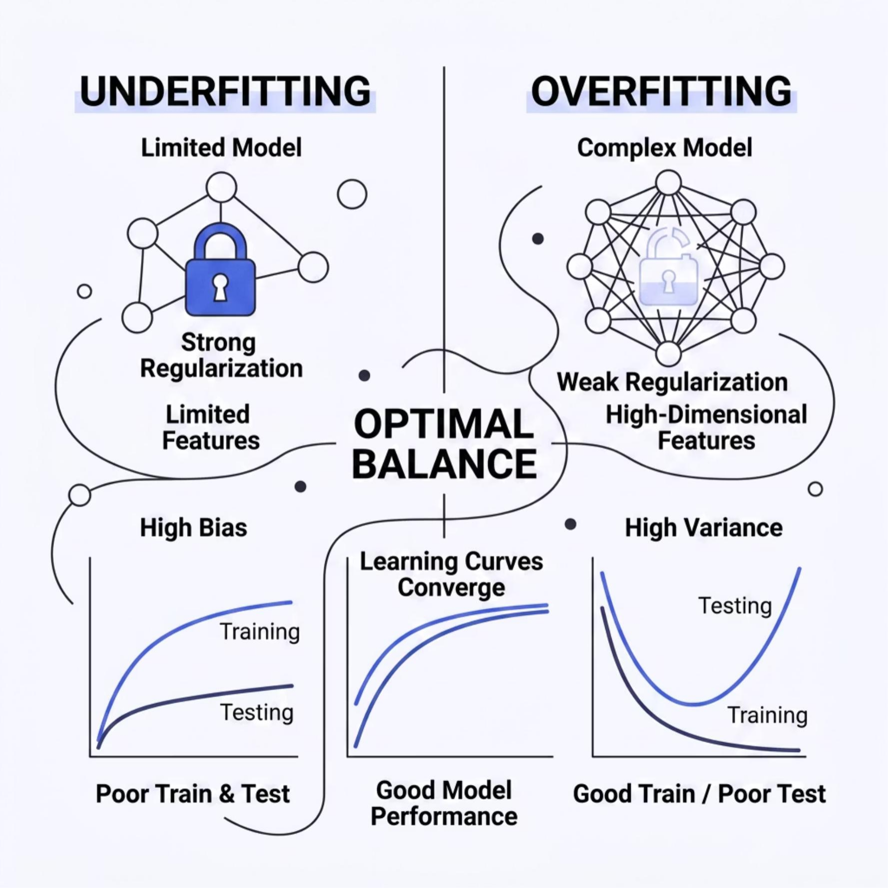
Harmonic mean of precision and recall for imbalanced data.

PR-AUC

Area under Precision-Recall curve; robust for skewed class distributions.

BIAS-VARIANCE TRADE-OFF

Overfitting & Underfitting Analysis



Learning curves and validation curves were used to diagnose model fit. The gap between training and validation performance reveals the degree of overfitting or underfitting across regularization strengths and feature subsets.

Key Insight: Optimal generalization occurs at intermediate regularization strength where training and validation curves converge.

REGULARIZATION STUDY

L1, L2 & Elastic Net

L1 – Lasso

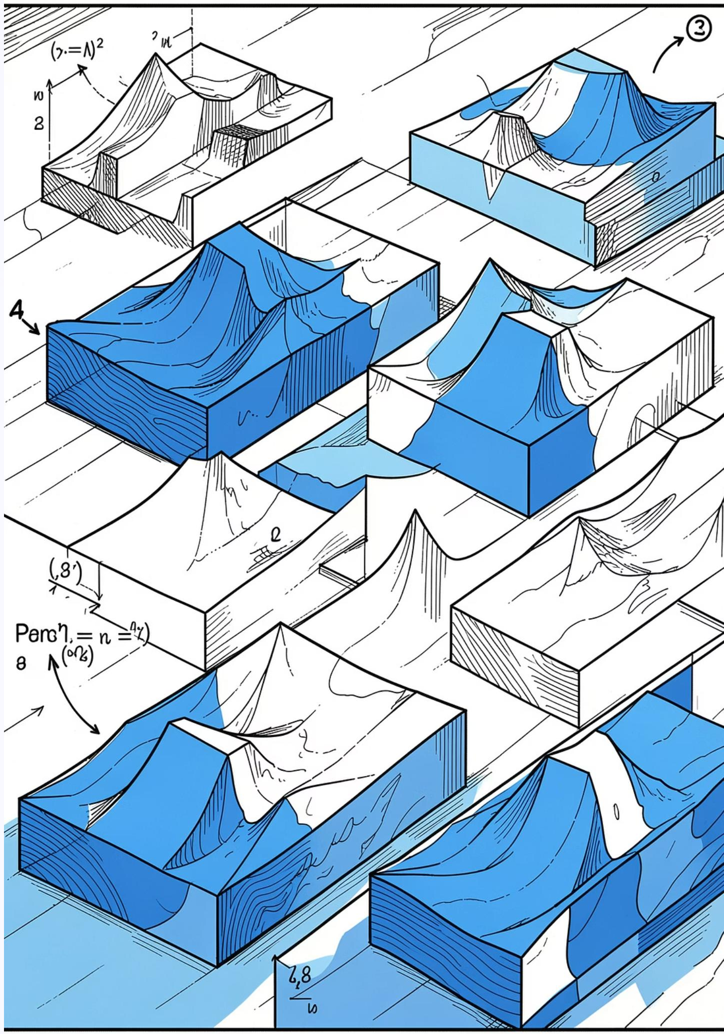
Induces sparsity by driving irrelevant coefficients to zero. Useful for feature selection in high-dimensional EEG data.

L2 – Ridge

Shrinks all coefficients proportionally, reducing variance without eliminating features. Improves stability.

Elastic Net

Combines L1 and L2 penalties, balancing sparsity with grouped feature selection for correlated EEG features.



CLASS IMBALANCE

Handling Imbalanced Seizure Classes

Strategies Evaluated

→ **SMOTE Oversampling**

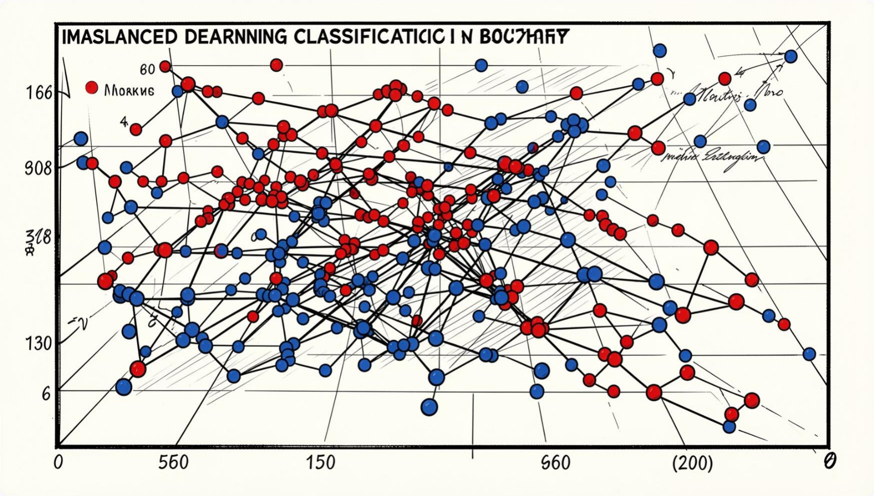
Synthetic minority examples generated to balance class distribution.

→ **Random Undersampling**

Majority class reduced to match minority class size.

→ **Class Weighting**

Higher penalty assigned to minority class misclassifications.



⚠ **Trade-off:** SMOTE improves recall for seizure events but may introduce synthetic noise affecting precision.

RESULTS

Experimental Findings

Preprocessing Order Matters

Pipeline sequencing significantly affected feature quality and downstream classification accuracy.

PCA Improved Stability

Dimensionality reduction via PCA enhanced feature stability and reduced overfitting risk.

Elastic Net Optimal

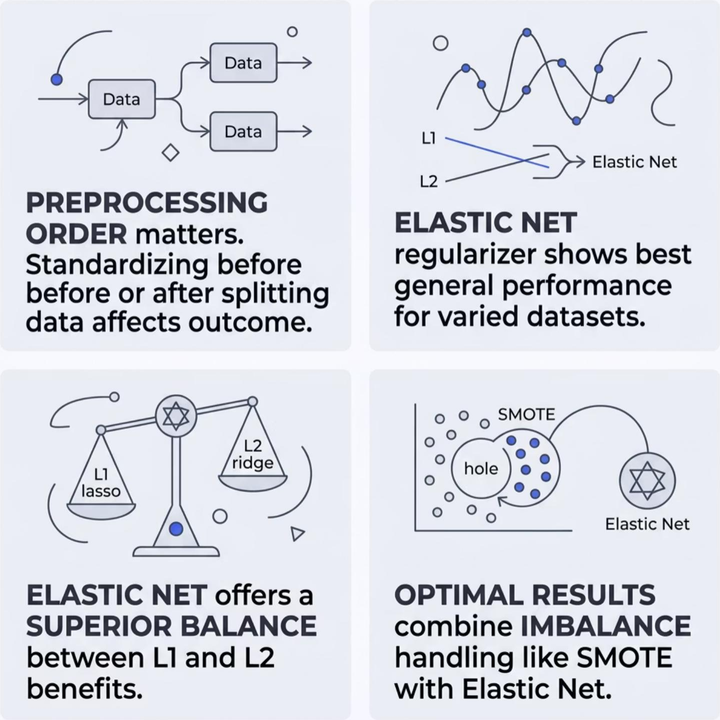
Elastic Net achieved the best balance between sparsity and coefficient stability across datasets.

SMOTE Boosted Detection

SMOTE oversampling improved minority (seizure) class detection rates without severe precision loss.

SYNTHESIS

Comparative Analysis & Conclusions



Key Conclusions

- Preprocessing order is a non-trivial design choice that directly influences model performance.
- Elastic Net consistently outperforms L1 and L2 in cross-dataset generalization.
- SMOTE combined with Elastic Net yields the strongest minority class detection.
- Learning curve analysis is essential for diagnosing optimal regularization strength.

Overall: Structured preprocessing + Elastic Net + SMOTE forms an effective pipeline for seizure prediction under class imbalance.